

Agentic AI Hackathon **2026**

AWS AI Services Introduction

AWS account setup, Kiro, Amazon Quick, cost guardrails and cleanup

Prashant Advani | Sr. Partner Solutions Architect | September 11, 2026



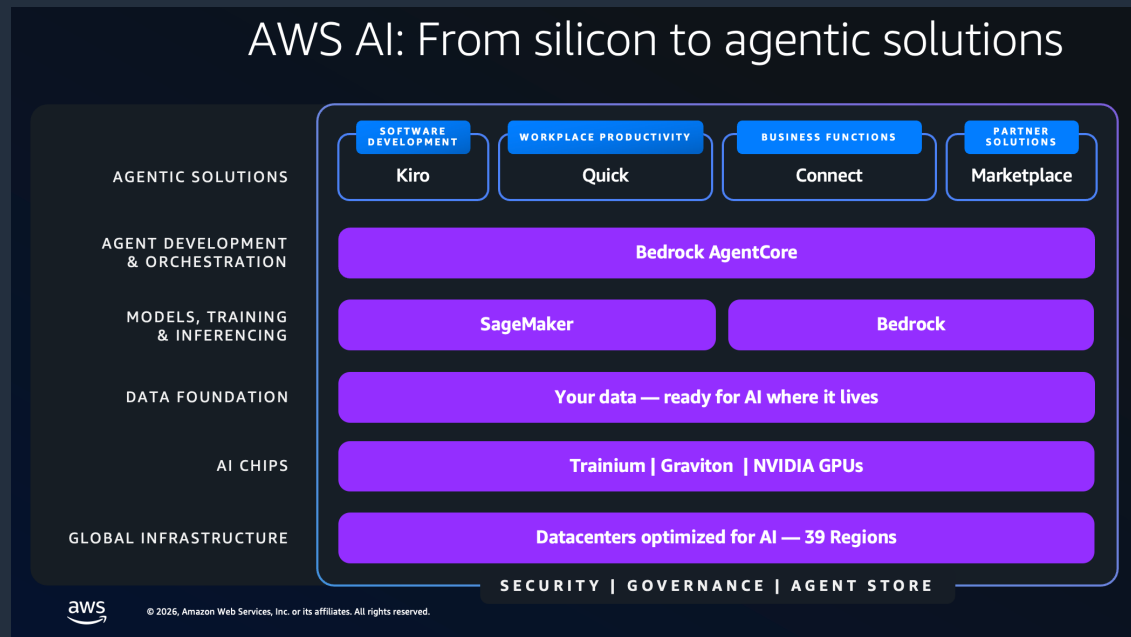
AI is changing every industry

The advantage comes from learning how to turn an idea into a safe, working system

During this hackathon you can

- Build a portfolio project for manufacturing or supply chain
- Use Kiro to move from requirements to working code
- Use Amazon Quick to research, analyze and create deliverables
- Learn AWS identity, cost control and cleanup habits
- Leave ready for the hands-on workshop and hackathon kickoff

AWS AI: From silicon to agentic solutions



Your goal today: leave ready to build and able to explain what you built.

Session plan

One hour from account setup to a clean shutdown

0–5 min

The three-account map

5–18 min

AWS account and security

18–30 min

Budget alerts and guardrails

30–42 min

Kiro features and safe AWS access

42–54 min

Amazon Quick setup and demo

54–60 min

Cleanup checklist and questions

Choose the setup path

Most students need the individual path for this hackathon

Student or individual path

AWS

Create a Free account plan when eligible. Protect root with MFA.

Kiro

Sign in with AWS Builder ID or a supported social login.

Amazon Quick

Create a standalone Free account with email or social login.

IAM Identity Center is optional for this path.

Organization-managed path

Use IAM Identity Center when an organization manages workforce access, Kiro subscriptions, Quick Professional or Enterprise, or several AWS accounts.

An administrator creates users or groups, permission sets and application assignments.

Free-plan warning

Creating or joining AWS Organizations automatically upgrades a Free account plan to the Paid plan and ends the Free-plan protection.

Three separate accounts

Use the same email if convenient, but keep the identities conceptually separate

AWS account

Cloud resources, billing, credits, IAM and service access.

Kiro identity

Sign in with AWS Builder ID, Google, GitHub or an organization identity.

Amazon Quick account

Fastest student path uses email or social login. An AWS account is optional.

Important

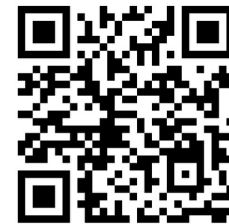
- A Kiro login does not automatically grant AWS permissions.
- A Quick Free account does not use the AWS bill.
- Only connect tools to the AWS account when the workshop needs it.



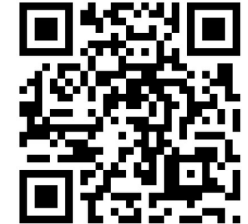
<https://builder.aws.com/start>



<https://aws.amazon.com/free/>



<https://kiro.dev/students/>



<https://aws.amazon.com/quick/>

Result: fewer sign-in surprises during the hands-on lab

AWS account plan

Choose the option that matches the workshop services and your risk tolerance

Free account plan

Best default for student experimentation when all required services are eligible.

- No charges while the Free plan is active
- USD 100 credits for eligible new accounts
- Up to six months, or until credits are used
- Some services and features are unavailable

Paid account plan

Use only when the workshop needs a service unavailable on the Free plan.

- Access to all AWS services and features
- Pay-as-you-go after credits
- Requires active budget controls
- Cleanup remains your responsibility

Workshop decision

Start with the Free plan unless the hands-on guide explicitly requires the Paid plan.

AWS account setup

Complete activation before the workshop starts

01 Email and account name

Use an email you can access throughout the hackathon. Verify the address.

02 Plan and contact details

Choose Free or Paid. Enter accurate contact information.

03 Billing verification

Add payment details if requested. A temporary verification hold may appear.

04 Phone verification

Complete the phone or text verification step.

05 Support and activation

Choose the available basic support option and wait for the activation email.

Bring a laptop with local administrator rights. Account activation can take up to 24 hours.

AWS Free account signup

Complete this before the event so activation time does not delay the workshop

1

Enter a root email you can keep using and a recognizable account name.

2

Verify the email, create a strong root password, and enter contact details.

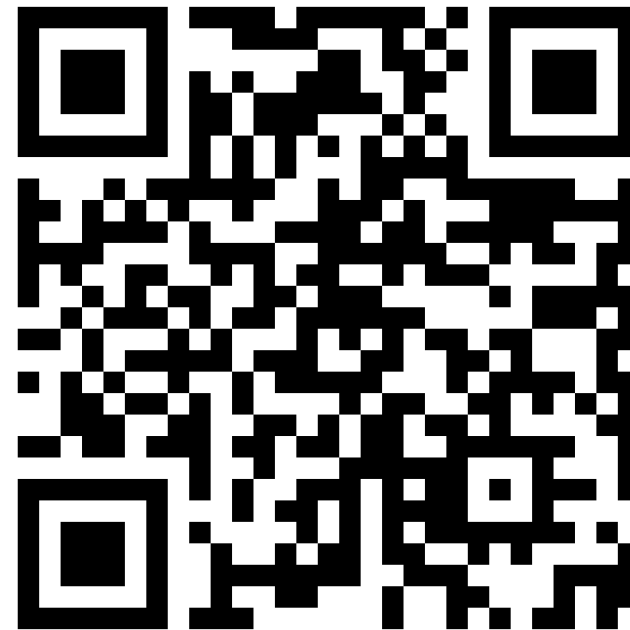
3

Choose the Free account plan when eligible. Add billing and phone verification if requested.

4

Wait for the activation email, then sign in and open Billing and Cost Management.

Never share the root password, payment details or MFA code.



aws.amazon.com/getting-started

Secure the AWS account on day one

Root user

Enable MFA immediately. Store the password securely. Use root only for tasks that require it.

Daily administrator

Create an administrative identity. Prefer IAM Identity Center where practical. Enable MFA.

Workload access

Use IAM roles and short-lived credentials. Do not paste long-lived access keys into code or chat.

Minimum security checklist

- Root MFA is active
- Daily sign-in uses a separate identity
- No credentials are committed to Git
- One team member owns billing notifications
- Account recovery email and phone remain available

Never share the root password or MFA code with teammates.

IAM Identity Center prerequisites

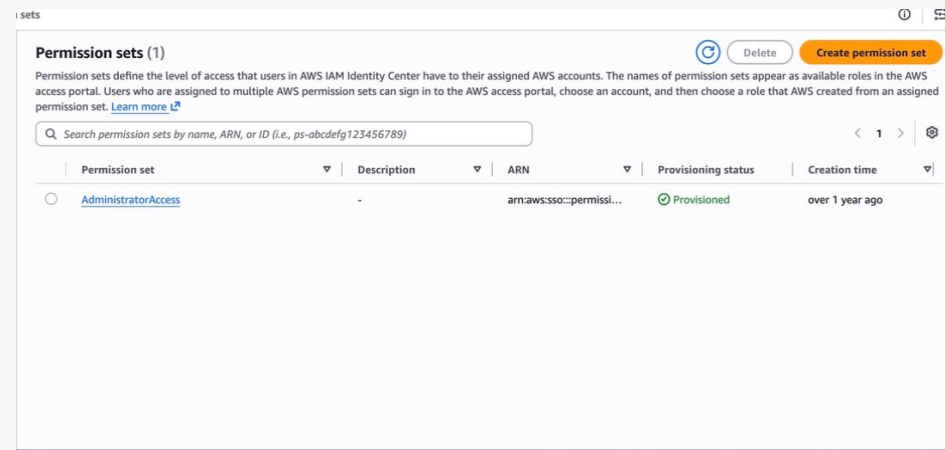
Use this managed path for organizational access to AWS accounts and applications

Before enabling

- Decide between an organization instance and a single-account instance
- Choose the Region carefully
- Confirm who administers users, groups and recovery
- Use an organization instance for several accounts
- Use an account instance for a simpler single-account application setup

Free-plan warning

Do not create AWS Organizations only to install Kiro or standalone Quick. That action upgrades the Free account plan.



Permission sets define what a workforce identity can do after signing in.

Console path: IAM Identity Center > Users and groups > Permission sets > AWS accounts or Applications

IAM Identity Center setup

The minimum managed setup for Kiro or Quick organization access

1

Enable an instance

Open IAM Identity Center in the chosen Region. Enable an account or organization instance.

2

Add people

Create a user and group, or connect the organization identity provider.

3

Create permissions

Create a permission set with only the AWS access needed for the workshop.

4

Assign access

Select the AWS account, user or group, and permission set. Submit the assignment.

5

Assign applications

For managed Kiro or Quick, assign the application or subscription to the intended group.

6

Test sign-in

Open the AWS access portal and verify the user sees only the expected account and applications.

sets > Create permission set

Select permission set type

A permission set contains policies that determine a user's permissions to access an AWS account. When you assign a user or group to a permission set in an AWS account, AWS IAM Identity Center creates an IAM role in the account and attaches the policies specified in the permission set to that role. Select an option to specify the permission set type. [Learn more](#)

Permission set type

Types

☒ Predefined permission set

Create a predefined permission set by choosing an AWS-defined template. This template enables you to select a single AWS managed policy. For example, you can select a policy that grants permissions for a common job function, such as Billing, or a specific level of access to AWS services and resources, such as ViewOnlyAccess. You can update the permission set as your needs evolve.

☐ Custom permission set

Create a custom permission set by selecting AWS managed policies and creating an inline policy (recommended). You can also attach customer managed policies and set a permissions boundary (advanced).

Policy for predefined permission set

Select an AWS managed policy

☒ AdministratorAccess

Provides full access to AWS services and resources.

☐ AmazonODBAutonomousDatabaseAdmin

Provides administrative access to manage Autonomous Database resources in Oracle Database@AWS.

Use the least-privilege policy that supports the workshop. Avoid AdministratorAccess for student accounts.

Use groups for repeatable access. Test with one student identity before inviting the whole cohort.

Project boundaries

Make every resource easy to find and remove

One Region

Use the workshop Region unless the instructions specify another. Avoid accidental resources in multiple Regions.

One naming prefix

Prefix resources with a team name, for example: team07-hackathon.

Three tags

Project=ID-Hackathon-2026
Team=Team07
DeleteAfter=2026-10-31

Why this matters

The same project prefix and tags let the team filter Cost Explorer, find resources in Tag Editor, and prove cleanup across Regions.

Create infrastructure with CloudFormation, CDK or another repeatable template whenever the workshop supports it.

Two budgets for the hackathon

Track the monthly run rate and the total project cap

Monthly budget

Resets each calendar month.

Use it to control the September and October run rate.

Custom-period budget

Covers the complete build window.

Suggested period: September 11 to October 31, 2026.

Recommended alert ladder

50%

Actual

Check the main cost drivers

80%

Actual

Stop nonessential experiments

100%

Actual

Freeze new resources and clean up

100%

Forecast

Useful after enough usage history exists

Set the amount to the organizer's cap or the maximum your team can safely spend.

Create the monthly cost budget

Billing and Cost Management > Budgets > Create budget

1

Choose Use a template and Monthly cost budget

2

Name it ID-Hackathon-Monthly

3

Enter the organizer's monthly cap

4

Add an email the team monitors

5

Create the budget and verify alerts

Create a second custom-period budget for 11 September to 31 October 2026.

Billing and Cost Management > Budgets > Create budget

Billing and Cost Management

Choose billing view [New](#)

Primary view

Home

Getting Started

Billing and Payments

Bills

Payments

Credits

Purchase Orders

Cost and Usage Analysis

Cost Explorer

Cost Explorer Saved Reports

Cost Anomaly Detection

Free Tier

Data Exports

Customer Carbon Footprint Tool

Cost Organization

Cost Categories

Cost Allocation Tags

Billing Conductor

Budgets and Planning

[Budgets](#) [New](#)

Budgets Reports

Pricing Calculator [New](#)

Savings and Commitments

Cost Optimization Hub [New](#)

▼ **Savings Plans**

Overview

Inventory

Recommendations

Purchase Analyzer [New](#)

Utilization Report

Coverage Report

Purchase Savings Plans

Choose budget type [Info](#)

Budget setup

☒ Use a template (simplified)
Use the recommended configurations. You can change some configuration options after the budget is created.

☐ Customize (advanced)
Customize a budget to set parameters specific to your use case. You can customize the time period, the start month, and specific accounts.

Templates - new
Choose a template that best matches your use case.

☐ Zero spend budget
Create a budget that notifies you once your spending exceeds \$0.01.

☒ Monthly cost budget
Create a monthly budget that notifies you if you exceed, or are forecasted to exceed, the budget amount.

☐ Daily Savings Plans coverage budget
Create a coverage budget for your Savings Plans that notifies you when you fall below the defined target.

☐ Daily reservation utilization budget
Create a utilization budget for your reservations that notifies you when you fall below the defined target.

Monthly cost budget - Template

Budget name
Provide a descriptive name for this budget.

My Monthly Cost Budget
Names must be between 1-100 characters.

Enter your budgeted amount (\$)
Last month's cost: \$0.00

100.00

Email recipients
Specify the email recipients you want to notify when the threshold has exceeded.

my@email-address.com
Maximum number of email recipients is 10.

Scope
All AWS services are in scope in this budget.

Alerts
You will be notified when 1) your actual spend reaches 85% 2) your actual spend reaches 100% 3) if your forecasted spend is expected to reach 100%.

Template settings
This template has default configurations that can be changed later. To change any of these settings, see [Custom](#). You can also download this template in [JSON](#).

[Cancel](#) [Create budget](#)

Budget alerts are not a real-time hard cap

AWS Budgets helps teams react, but spend can continue between data updates.

Alert

Email at 50%, 80% and 100% actual spend. Keep the recipient monitored.

Restrict

At 90% or 100%, a Budget Action can attach a deny policy that blocks new resources.

Stop

A Budget Action can target selected EC2 or RDS resources. It does not cover every service.

Strongest beginner guardrail: Free account plan + small architecture + alerts + cleanup.

Kiro

Agentic AI development from prototype to production

Kiro combines an AI coding assistant with a structured development workflow.

Vibe mode

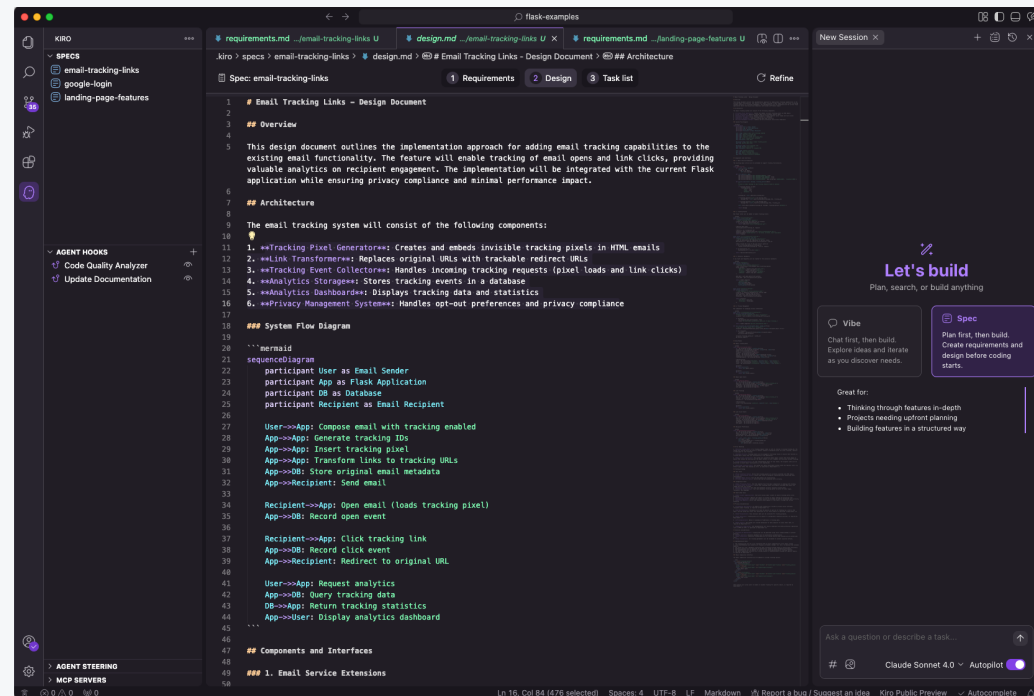
Explore ideas and iterate quickly.

Spec mode

Create requirements, design, and tasks before coding.

Project context

Use steering, hooks, and MCP to guide repeatable work.



Student starting point: use one small feature to compare Vibe and Spec modes.

Kiro: first project and core features

A short preview before the hands-on Kiro workshop

01 Setup

Download from kiro.dev, sign in, and open a small project folder.

02 Chat

Ask Kiro to explain the codebase, propose a change, or help debug.
Review every edit.

03 Specs

Turn an idea into requirements, design, and tasks before implementation.

04 Steering, hooks and MCP

Keep project context, automate routine checks, and connect approved tools.

First exercise: create one small spec and complete only the first task. The workshop will go deeper.

Connect Kiro to AWS safely

1. Configure AWS access

Use the workshop's supported method:

- IAM Identity Center profile
- Temporary lab credentials
- A local AWS CLI profile

Avoid root credentials.

2. Confirm identity

Run a read-only identity check before deploying:

```
aws sts get-caller-identity
```

Confirm the account ID and role match the team account.

3. Limit tool approval

Review every command Kiro proposes. Start with read-only actions. Do not auto-approve destructive or high-cost operations.

Keep secrets out of prompts, screenshots, repositories and shared chat.

Amazon Quick

An AI assistant that turns questions into answers and actions

Quick brings research, team knowledge, agents, workflows, apps, and deliverables into one place.

Research

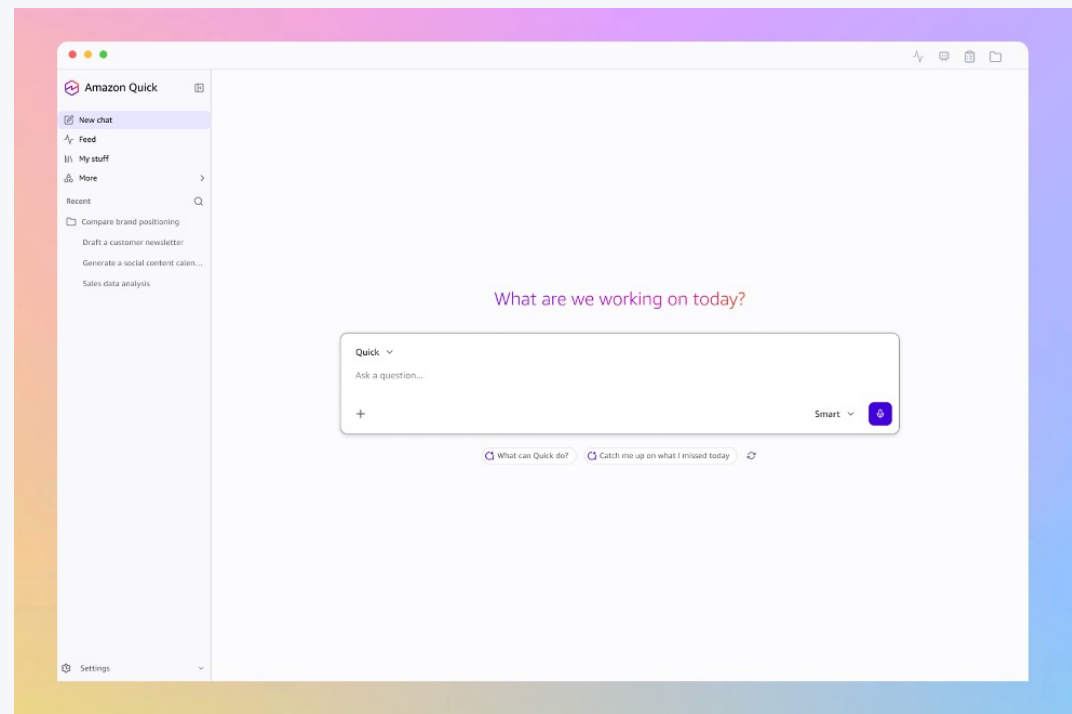
Create cited reports from public sources and team context.

Spaces and agents

Organize shared knowledge and build assistants for repeatable work.

Actions and outcomes

Automate tasks, analyze data, and create finished deliverables.



Student starting point: use Research to turn a broad idea into a cited problem brief.

Amazon Quick: setup and feature map

The standalone Free account is the fastest student path

Start in two minutes

1. Open aws.com/quick
2. Create a standalone Free account
3. Start in Chat or open Research
4. Create a Space for the team
5. Upload only safe, non-sensitive files

No AWS account or credit card is required.

Features to try

Chat

Ask questions and create first drafts.

Research

Generate a cited report from public sources.

Spaces and agents

Keep project context, build custom agents, and connect approved tools.

New standalone accounts include a 30-day Plus trial. Choose Free afterward unless the team intends to pay.

Amazon Quick demo

Turn a broad idea into a cited problem brief in five minutes

Demo prompt

Research common causes of supplier delivery delays in Indonesian pulp and paper manufacturing.

Use public sources. Create a one-page brief with:

- assumptions and evidence
- the people affected
- three prototype opportunities

Demo flow

1. Open Research and review the plan
2. Run the report and inspect citations
3. Ask Quick for a problem statement
4. Save the work in a team Space
5. Export the brief for the pitch

Keep a completed report ready in case live research takes longer than the session.

Cost-aware build defaults

Use one Region

Reduce forgotten resources and simplify troubleshooting.

Prefer on-demand

Avoid commitments and provisioned capacity for a short hackathon.

Use serverless first

Pay for requests and execution time instead of idle servers.

Keep datasets small

Synthetic or sampled data is enough for a working prototype.

Turn off idle compute

Stop or delete notebooks, instances, databases and endpoints.

Review daily

Check Cost Explorer and the budget dashboard at the end of each build day.

Avoid NAT gateways, large GPU instances, persistent endpoints and high-volume logging unless the design truly needs them.

Cleanup sequence

Delete in reverse order of creation

1

Delete application entry points

API Gateway APIs, load balancers, CloudFront and public endpoints.

2

Delete agent resources

AgentCore runtimes, gateways, targets, memory, Bedrock agents and knowledge bases.

3

Delete compute and data

Lambda, containers, EC2, RDS, DynamoDB, OpenSearch and SageMaker endpoints.

4

Delete storage and images

Empty S3 buckets first. Remove ECR images, EBS volumes and snapshots.

5

Delete infrastructure stacks

Delete CloudFormation or CDK stacks and resolve any retained resources.



<https://aws.amazon.com/getting-started/>

Do not delete shared or organizer-provided resources. Confirm ownership before cleanup.

Cleanup audit

A deletion click is not proof that spending stopped

Resource search

Search the project tag across all relevant Regions.

Use Resource Explorer for untagged resources with:

tag:none

Billing check

Review Cost Explorer and the Budgets dashboard after cleanup.

Remember that billing data can lag.

Tool accounts

Quick: choose Free after the Plus trial unless you intend to pay.

Kiro: sign out on shared machines and remove unused local credentials.

Final team sign-off

- No project resources remain in the workshop Region
- No unexpected resources appear in other Regions
- The latest cost trend is flat
- Paid trials or subscriptions are intentional

Ready for the workshop

- AWS account activated and root MFA enabled
- Monthly and hackathon-period budgets created
- Kiro installed and signed in
- Quick account created using the intended plan
- Laptop has administrator rights
- Team naming prefix, Region and tags agreed

Official references

AWS account
docs.aws.amazon.com/accounts/

AWS Budgets
docs.aws.amazon.com/cost-management/

Kiro
kiro.dev

Amazon Quick
docs.aws.amazon.com/quick/

Questions

Terima kasih

Mari kita bangun sesuatu yang hebat!

Terima kasih atas waktu dan partisipasinya.

Prashant Advani – padvani@amazon.com



<https://www.linkedin.com/in/prashantadvani/>

LET'S GO BUILD SOMETHING GREAT!